

Resolution of the Slow-Rotating Magnetar Paradox via QCD Vacuum Permeability Phase Transitions

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Repository: <https://github.com/infosave2007/vmf>

Abstract

The origin of the extreme magnetic fields ($B \sim 10^{14} - 10^{15}$ G) observed in magnetars is traditionally attributed to an active $\alpha - \omega$ dynamo mechanism operating during the first tens of seconds of a proto-neutron star's life. However, this turbulent dynamo strictly requires initial birth spin periods on the order of $P \sim 1 - 3$ ms. The discovery of highly magnetized neutron stars with inferred birth periods significantly larger than this threshold, alongside ultra-long period magnetars (e.g., 1E 161348-5055), poses a severe challenge to standard dynamo theory. We present an analytic, non-parametric resolution derived from the Null-Vector Gravity (NVG) framework. We demonstrate that the macroscopic QCD vacuum condensate, which couples to baryonic mass, undergoes a structural phase transition at densities exceeding $\rho_c \approx 7 \times 10^4$ MeV/fm³. The density-driven “melting” of the vacuum mass fraction ($\epsilon \approx 0.91$) induces a proportionate collapse in the local vacuum magnetic permeability within the stellar core. Conserving magnetic flux across this phase boundary results in a deterministic, volume-averaged quantum amplification of the core magnetic field by a factor of $\langle \Gamma_{mag} \rangle \approx 7.4$, completely independent of the star's rotation rate. This structural amplification mechanism eliminates the requirement for millisecond birth periods, naturally explaining the existence of highly magnetized, slow-rotating compact objects.

1. Introduction

Magnetars are a class of isolated neutron stars characterized by extreme magnetic fields ($10^{14} - 10^{15}$ G), manifesting observationally as Soft Gamma Repeaters (SGRs) and Anomalous X-ray Pulsars (AXPs). In the standard astrophysical paradigm, first formalized by Duncan and Thompson (1992), these colossal fields are not fossil fields inherited from the progenitor star, but are generated via a vigorous convective $\alpha - \omega$ dynamo in the hot proto-neutron star phase.

A fundamental requirement for this dynamo to operate efficiently is that the star must be born rotating at near-breakup speeds, with an initial spin period $P_i \sim 1 - 3$ ms. If a star is born rotating more slowly ($P_i \geq 5$ ms), the dynamo fails, and a standard radio pulsar ($B \sim 10^{12}$ G) is formed.

However, multi-wavelength observations have increasingly contradicted this requirement. Energetic studies of supernova remnants associated with magnetars show no evidence of the massive rotational energy injection ($\sim 10^{52}$ ergs) expected from a millisecond braking proto-neutron star. Furthermore, the discovery of magnetars in young remnants with inferred slow birth spins, and extreme outliers like the 6.67-hour period source 1E 161348-5055, suggest that the dynamo mechanism is either incomplete or entirely unnecessary.

In this letter, we apply the Null-Vector Gravity (NVG) framework to propose a purely structural, rotation-independent mechanism for magnetic field amplification during core collapse.

2. The Vacuum Mass Fraction Phase Transition

In the NVG model, the invariant mass of a nucleon is not a static property but a dynamic coupling to a background macroscopic Quantum Chromodynamics (QCD) condensate. The bare mass m_0 contributes only $\sim 9\%$, while the structural vacuum mass fraction $\epsilon = m_s/m_N \approx 0.91$ is anchored by the $M_{\Omega,0} = 859$ MeV mass gap.

During the core collapse of a massive star, baryon density n_B rapidly exceeds nuclear saturation density. As the local energy density approaches the energy density of the vacuum condensate itself ($\rho_c \approx 7 \times 10^4$ MeV/fm³), a dielectric screening effect is initiated. The structural mass fraction ϵ undergoes a continuous melting phase transition:

$$m_{eff}(n_B) = m_N \left(1 - \epsilon \left[1 - \exp \left(-\frac{n_B}{n_{crit}} \right) \right] \right)$$

While this mechanism successfully stiffens the Equation of State (EOS) to support $2.25M_\odot$ neutron stars, it also has profound implications for the electromagnetic properties of the local vacuum.

3. Magnetic Amplification via Permeability Collapse

In classical electrodynamics, the vacuum magnetic permeability μ_0 is treated as an immutable universal constant. However, in the NVG framework, the vacuum is a highly structured, non-perturbative medium. The same QCD condensate that imparts structural mass to fermions also defines the local electromagnetic permeability.

As the dense matter core undergoes the VMF phase transition, the melting of the structural mass fraction ϵ implies a breakdown of the macroscopic vacuum structure. Consequently, the local effective permeability μ_{eff} must scale proportionally to the surviving vacuum structure:

$$\mu_{eff}(n_B) \approx \mu_0 (1 - \epsilon_{melted}(n_B))$$

At the center of a $1.4M_\odot$ neutron star, where the melting is nearly complete ($\epsilon_{melted} \rightarrow 0.91$), the effective permeability drops by an order of magnitude: $\mu_{eff} \rightarrow 0.09\mu_0$.

According to the laws of electrodynamics, the total magnetic flux $\Phi_B = \int \vec{B} \cdot d\vec{A}$ must be conserved during the collapse. Because the constitutive relation linking the auxiliary magnetic field $\vec{H}$ and the physical magnetic field $\vec{B}$ is defined by $\vec{B} = \mu_{eff}\vec{H}$, a drop in μ_{eff} forces the internal energy of the pre-existing fossil magnetic field to undergo a structural quantum amplification in the frame of the baryonic matter.

Integrating this effect over the radial density profile of a canonical $1.4M_\odot$ TOV core yields a volume-averaged magnetic amplification factor:

$$\langle \Gamma_{mag} \rangle = \frac{1}{V} \int_0^R \frac{1}{1 - \epsilon_{melted}(r)} 4\pi r^2 dr \approx 7.4$$

4. Resolving the Dynamo Paradox

This derivation provides a direct resolution to the magnetar birth period paradox. When a progenitor star with a strong fossil magnetic field undergoes core collapse, the resulting magnetic field of the neutron star is traditionally calculated assuming flux conservation in a constant μ_0 vacuum, yielding $B \sim 10^{12} - 10^{13}$ G.

However, by accounting for the NVG vacuum permeability phase transition, the true field strength is automatically multiplied by $\langle \Gamma_{mag} \rangle \approx 7.4$. Because magnetic energy scales as B^2 , this represents a ~ 55 -fold increase in the magnetic energy density of the core, easily pushing a high-end fossil field (5×10^{13} G) deep into the magnetar regime (3.7×10^{14} G).

Crucially, this amplification is driven entirely by the scalar baryon density n_B and the static QCD phase transition. It requires zero rotational kinetic energy and is completely independent of the star's spin period P_i .

5. Conclusion

The standard $\alpha - \omega$ dynamo model relies on the assumption that the vacuum interior of a neutron star behaves electromagnetically identically to the vacuum

of empty space. The Null-Vector Gravity framework demonstrates that this assumption is flawed at supra-nuclear densities.

By recognizing that the QCD mass gap anchors both the structural mass of matter and the permeability of the local vacuum, we derive a density-driven magnetic amplification factor of $\langle \Gamma_{mag} \rangle \approx 7.4$. This provides a zero-parameter physical mechanism for generating magnetar-level fields from standard fossil fields, eliminating the need for millisecond birth periods. The NVG model naturally accommodates the existence of highly magnetized, slow-rotating compact objects, offering a rigorous, testable alternative to classical dynamo theory.

Data Availability

Full analytic derivations and Python computational suites calculating the TOV density profiles and radial permeability integration are publicly available at the NVG repository: [<https://github.com/infosave2007/vmf>] and archived on Zenodo [<https://zenodo.org/records/20214457>].

References

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